Overview of Kalman Filter Theory with Applications to Navigation Square Root Information Filters

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October 8, 2018

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Square Root Information Filters

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- When S is "large" P is "small" (in norm)

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- Substitute the Kalman gain formula $K = P^- H^\top \left[HP^- H^\top + R\right]^{-1}$ into the above equation to yield $P^+ = P^- - P^- H^\top \left[HP^- H^\top + R\right]^{-1} HP^-$

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- To futher expand and then simplify, we need to apply the Matrix Inversion Lemma (also known as the Woodbury formula): $(A + BCD)^{-1} = A^{-1} - A^{-1}B\left(C^{-1} + DA^{-1}B\right)^{-1} DA^{-1}$

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- We obtain: $[P^+]^{-1} = [P^-]^{-1} + H^\top R^{-1} H$, or, in terms of S: $S^+ = S^- + H^\top R^{-1} H$

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- The gain comutation takes on a very simple form: $K = S^{+} H^{\top} R^{-1}$
- The state is then updated using the gain in the normal way
- Standard KF requires inversion of $r \times r$ matrix (r =number of measurements); information filter requires inversion of a couple of $n \times n$ inversions.
  Thus, if $r >> n$, it is more efficient to use the information filter.

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$$S^{-}(k) = Q^{-1} - Q^{-1}\Phi\left[S^{+}(k-1) + \Phi^{\top}Q^{-1}\Phi\right]^{-1}\Phi^{\top}Q^{-1}$$

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- If the process noise covariance Q and/or the measurement noise covariance are constant (or change only infrequently) with time, then the indicated inverses can be computed once at startup or perhaps only occasionaly.

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- Conditioning refers to the condition number of a matrix:
  $\kappa(P) := \sigma_{\text{max}}/\sigma_{\text{min}}$,
  where σ_{max} is the maximum singular value of P and σ_{min} is the minimum singular value.
  Singular values of real-valued P are square roots of eigenvalues of $P^\top P$ (or $PP^\top$)

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- Note that $\kappa(P) \geq 1$, for a nonsingular matrix P, since all of its singular values are positive

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- Example of 2-state model: Highly uncertain first state, but highly certain second state:
$$P = \begin{bmatrix} 10^{10} & 0 \\ 0 & 10^{-10} \end{bmatrix}$$

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- MATLAB Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND = 1.000000e-20

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- If we can factor P as the product of a matrix and its transpose, i.e., $P = WW^\top$, then the matrix W is said to be the matrix square root of P

  Note: There are several alternative definitions in the literature $WW^\top, W^2, W^\top W$

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  T is **orthogonal** if its transpose is its inverse:
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- If $P = W_1 W_1^\top = W_2 W_2^\top$, then there is an orthogonal matrix T such that $W_2 = W_1 T$

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Square Root Filters: Orthogonal Matrices

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  - Length Invariance (Isometry): $\|Tx\| = \|x\|$
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  - Standard Gaussian Invariance: If v is zero-mean Gaussian random vector such that $\mathrm{E}\left[vv^\top\right] = I$, then for orthogonal T the vector Tv is also zero-mean Gaussian and $\mathrm{E}\left[Tv(Tv)^\top\right] = I$

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- Length and Standard Gaussian Invariances play crucial role in square root filtering